

Comparing prompt variants for EEG report classification

We evaluated **four prompt variants** for MedGemma-27B, each on both datasets and both quantizations (**4 prompts × 2 datasets × 2 quants = 16 runs**; all numbers are pooled over **1994 reports** — 1495 Zoe + 499 Maria — scored against annotator LD). The baseline model already matched the paper’s Mistral-7B everywhere except one class; these prompts are the attempt to close that gap.

Prompt	What it is
v1	Original baseline — Impression-first, short definitions.
v2	Professor’s revision — neurologist system role, read the report <i>body</i> first, keep body findings even when the Impression is conservative, extended clinical definitions.
v3	Ours — v1 + explicit Focal-Epi exclusions (don’t count generalized discharges, benign variants, artifacts, or focal slowing as focal epileptiform).
v4	Ours — v1 + a structured procedure (detect epileptiform → localize focal vs generalized → assign).

Bottom line: v3 is the best variant (1679 / 1994 reports fully correct). It fixes the one weak class — **Focal Epi 0.80 → 0.83, matching Mistral-7B** and beating it on Maria (0.90 vs 0.81) — without hurting the rest. v2 (the professor’s) is the weakest overall because it pushes Focal Epi the wrong way.

Why v3 wins: it starts from v1 (already our strongest baseline — 4 of 5 categories at or above Mistral) and changes exactly one thing — explicit exclusions for the one weak class, Focal Epi. That single, targeted fix lifts Focal Epi (0.80 → 0.83, via higher precision) and even nudges Gen Non-epi up, while leaving the other categories essentially untouched — so it keeps every strength and repairs the only weakness, for a net +7 fully-correct over v1. v4 targets the same class with a step-by-step procedure but the plain exclusion list works slightly better; v2 pushed the opposite way and broke Focal Epi.

1. Overall — which prompt is best

What you see: reports where all five labels are correct. v3 (1679) edges out the v1 baseline (1672); v4 is close (1662); **v2 is clearly weakest (1576)**. The spread is small because four of five categories barely move between prompts — the real differences are in one place (next chart).

2. Where the variants actually differ

What you see: on **Abnormality, Gen Epi, Focal Non-epi** all four prompts are within a point or two — the prompt hardly matters there. The one category that separates them is **Focal Epi**: v2 collapses (0.64), the baseline v1 sits at 0.80, and our **v3/v4 lift it to 0.83/0.82**. v3/v4 also nudge **Gen Non-epi** up (0.84 → 0.86–0.87).

3. The target — Focal Epi

What you see: Focal Epi F1 by prompt, per dataset and pooled, with Mistral-7B for reference. v3/v4 reach Mistral’s pooled level (0.83), and on **Maria they beat it** (0.90 / 0.89 vs 0.81). v2 is far below.

What you see (why it works): Focal Epi is precision-limited — the model catches almost every true case (recall ~0.97) but over-calls. v3/v4 raise **precision** (0.68 → 0.73) by cutting false alarms while

MedGemma-27B — reports fully correct by prompt variant (Q2_K)

v3 is the new best; v2 is the weakest

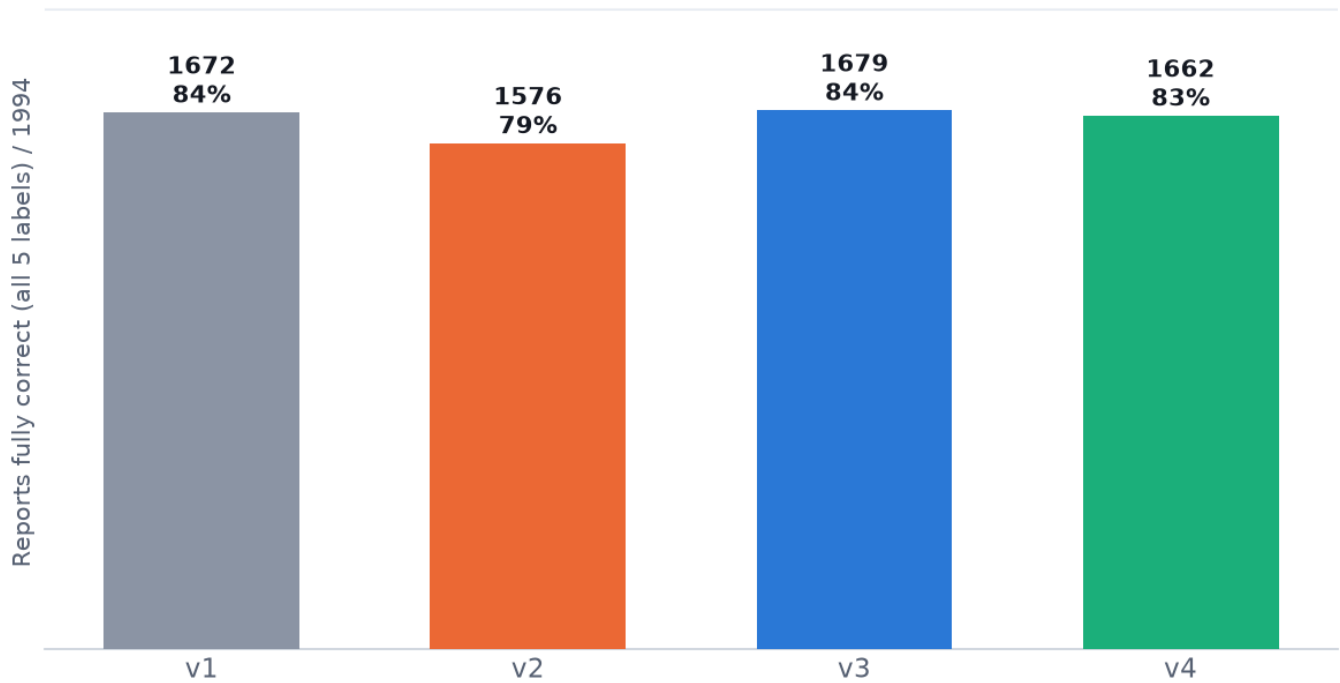


Figure 1: Fully correct by variant

MedGemma-27B — all four prompt variants, every category (pooled, Q2_K)

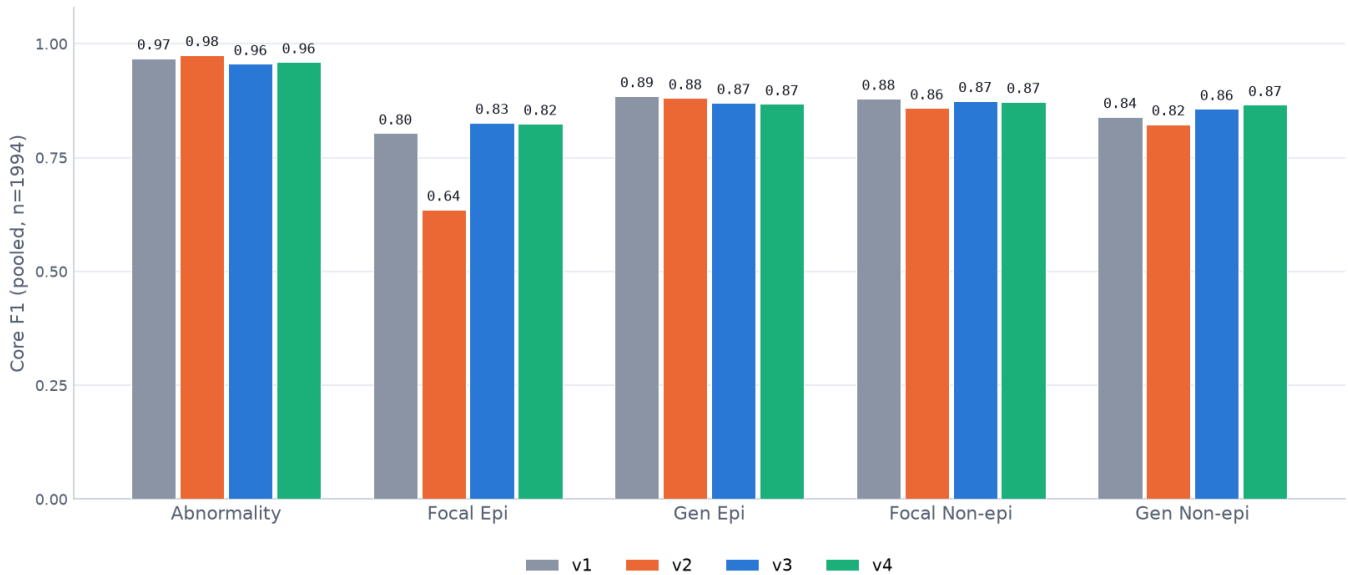


Figure 2: All variants by category

MedGemma-27B — Focal Epi F1 by prompt version (Q2_K)

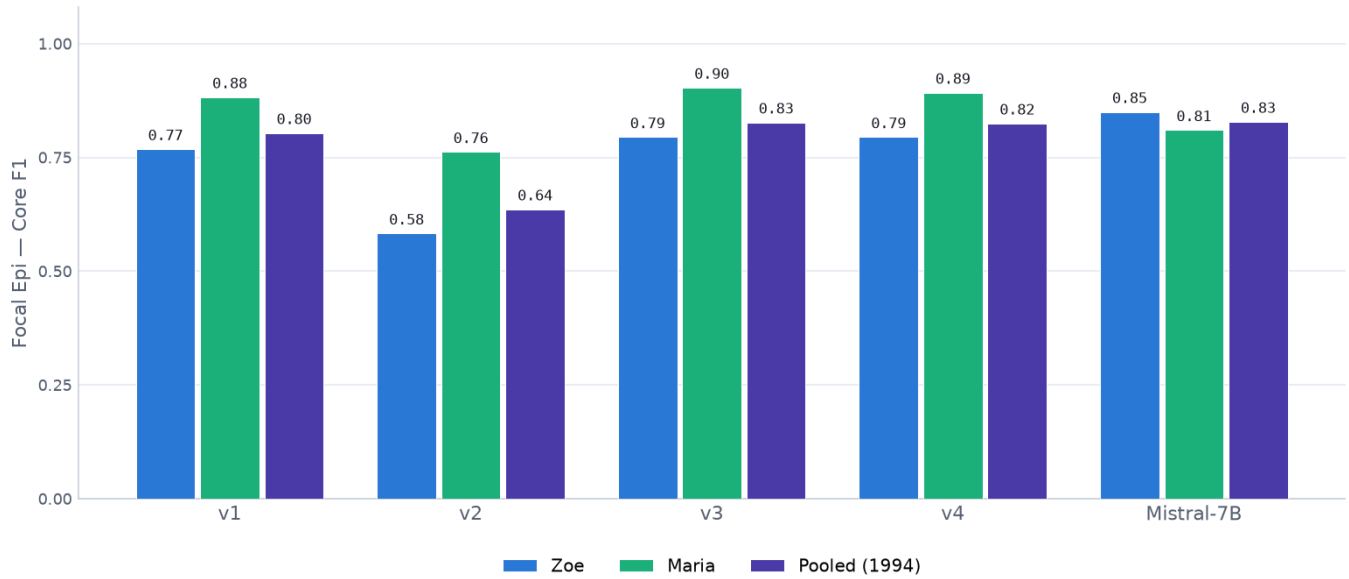


Figure 3: Focal Epi by prompt

Why v3/v4 help: they raise Focal Epi precision, keeping recall

v2 over-calls (precision crashes); v3/v4 cut false alarms

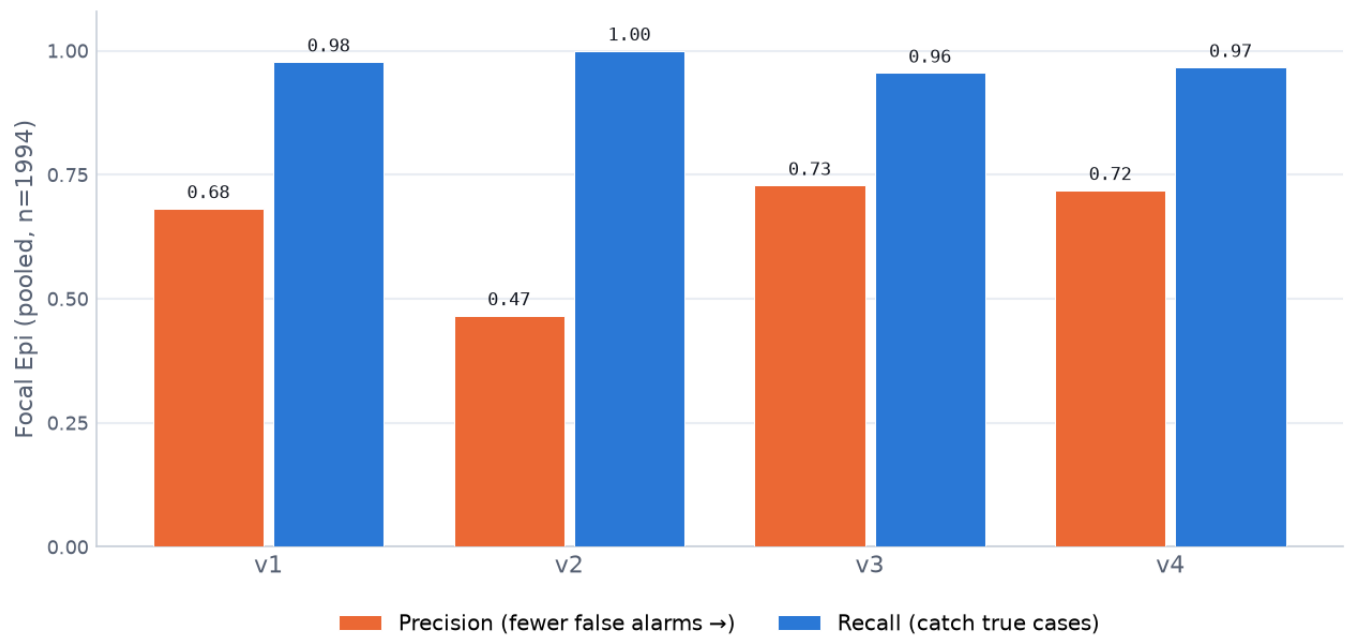


Figure 4: Focal Epi precision/recall

keeping recall; **v2 does the opposite** (precision 0.47). Making the model *more conservative* about focal epileptiform is the correct fix, which is the reverse of what v2 encouraged.

Numbers (Q2_K, pooled Core F1 vs LD)

Prompt	Abnorm	Focal Epi	Gen Epi	Focal Non	Gen Non	fully-correct
v1	0.97	0.80	0.89	0.88	0.84	1672
v2	0.98	0.64	0.88	0.86	0.82	1576
v3	0.96	0.83	0.87	0.87	0.86	1679
v4	0.96	0.82	0.87	0.87	0.87	1662
Mistral-7B	0.95	0.83	0.75	0.76	0.75	1485

Q2_K beats Q4_K_S for every prompt, so **v3 + Q2_K is the recommended configuration**.

Note on the charts above: all the variant comparisons use the **Q2_K** quantization (the recommended one). The quantization comparison is below.

4. Quantization — Q2_K vs Q4_K_S, per prompt

MedGemma-27B — Q2_K vs Q4_K_S, by prompt

Q2 wins for v1/v3/v4; Q4 only helps v2 (y-axis starts at 1400)

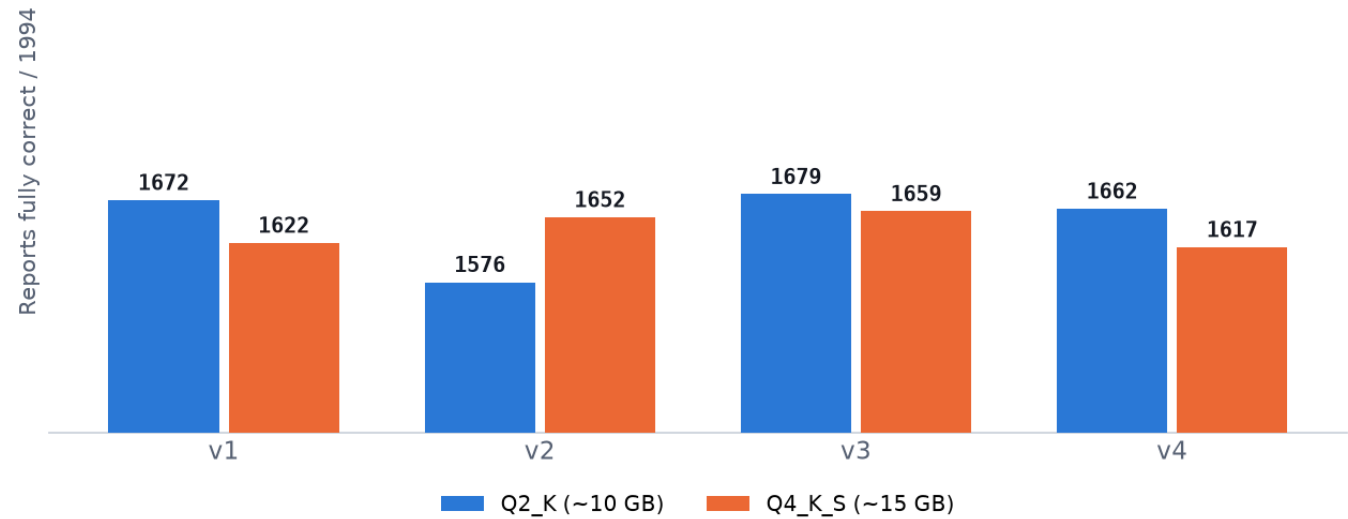


Figure 5: Q2 vs Q4 by prompt

What you see: on whole-report accuracy the smaller **Q2_K is ahead for v1, v3, and v4**; the larger Q4_K_S is only ahead for v2. Since Q4_K_S is ~50% bigger with no whole-report gain on the stronger prompts, we use **Q2_K as the default**. This is also why the cross-model comparison (vs Mistral, vs human) is done at **v1 + Q2_K**: v1 is the plain baseline prompt, and there Q2_K is higher on Zoe and level on Maria, at half the size — the larger quant buys nothing on the clean prompt.

Broken down by category:

MedGemma-27B — Q2_K vs Q4_K_S, per prompt and category (pooled n=1994, Core F1; y starts at 0.5)

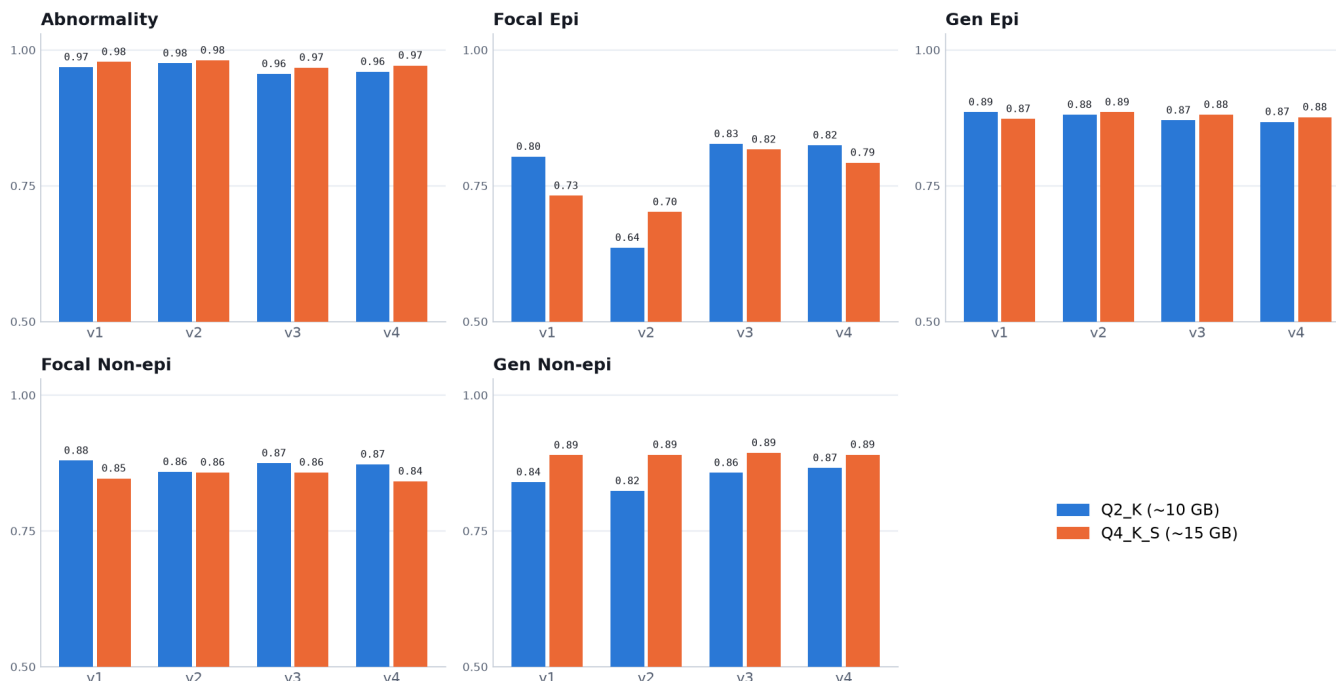


Figure 6: Q2 vs Q4 per prompt and category

What you see: the quantization effect is category-dependent and small. Per category the two are close, with **Q4_K_S a little higher on 3 of 5** (averaged over the four prompts):

Category	Q2_K	Q4_K_S	Δ (Q4–Q2)
Abnormality	0.965	0.975	+0.010
Focal Epi	0.773	0.761	−0.012
Gen Epi	0.876	0.879	+0.003
Focal Non-epi	0.871	0.851	−0.020
Gen Non-epi	0.847	0.891	+0.044

So per category Q4 often looks a touch higher. But **whole-report** accuracy (all 5 labels right) leans the other way:

Prompt	Q2_K fully-correct	Q4_K_S fully-correct
v1	1672	1622
v2	1576	1652
v3	1679	1659
v4	1662	1617

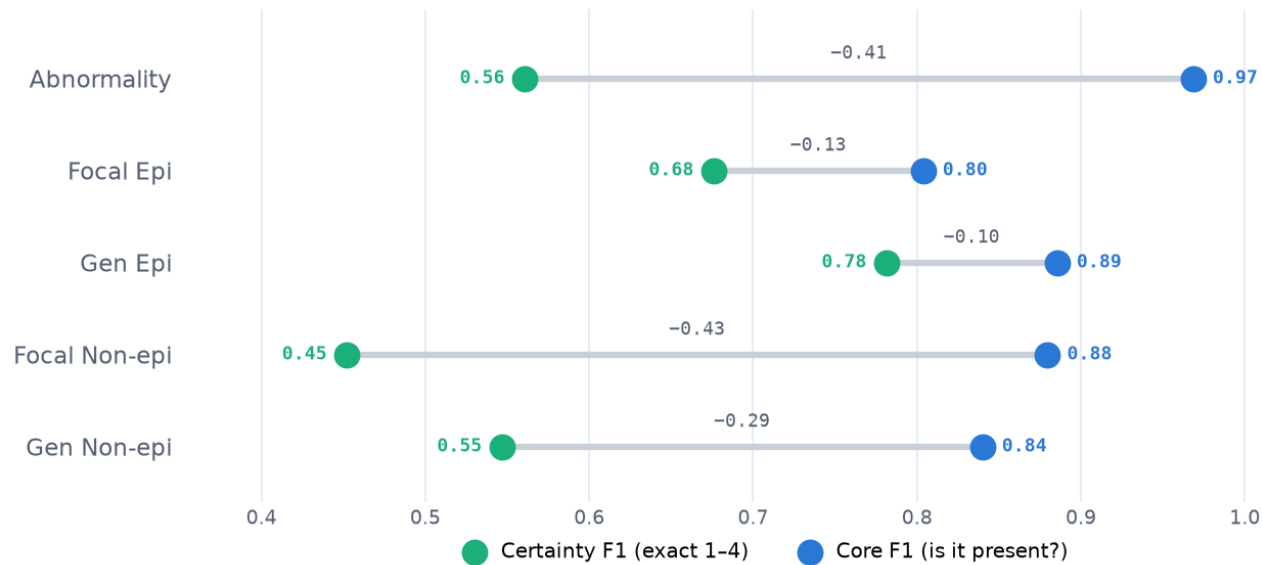
Why the two views disagree: Q2_K is higher on **Focal Non-epi**, a *common* class (~27% of reports), so its −0.02 under Q4 breaks many whole reports; Q4’s larger Gen Non-epi edge often doesn’t flip a report to fully-correct because it still fails another label. The gaps are small either way — we default to Q2_K for whole-report accuracy and half the size, while Q4_K_S is a little stronger if one specific class like Gen Non-epi matters most.

Where the prompts live

All four are in [core/prompt.py](#): PROMPT_PREFIX (v1), PROMPT_PREFIX_V2/V3/V4, and SYSTEM_V1/V2; selected by the PROMPT_VARIANT env var. Print any one, e.g.: `python -c "import core.prompt as p; print(p.PROMPT_PREFIX_V3)"`.

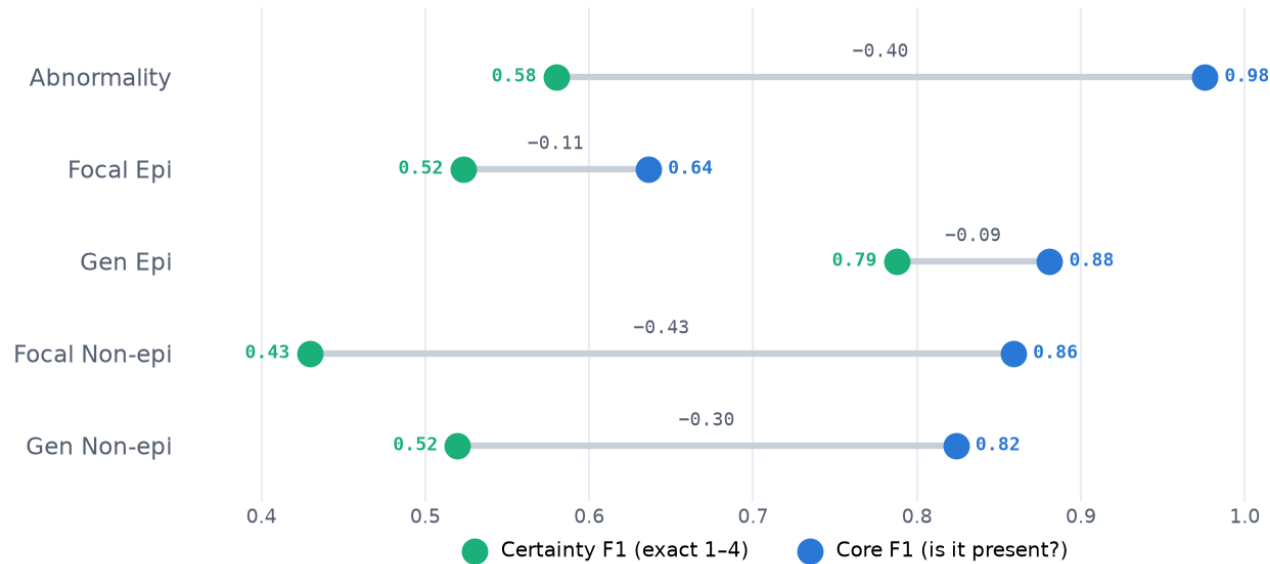
MedGemma-27B · v1 · Q2_K — core vs certainty

F1: binary present/absent vs strict exact-level match · pooled n=1994



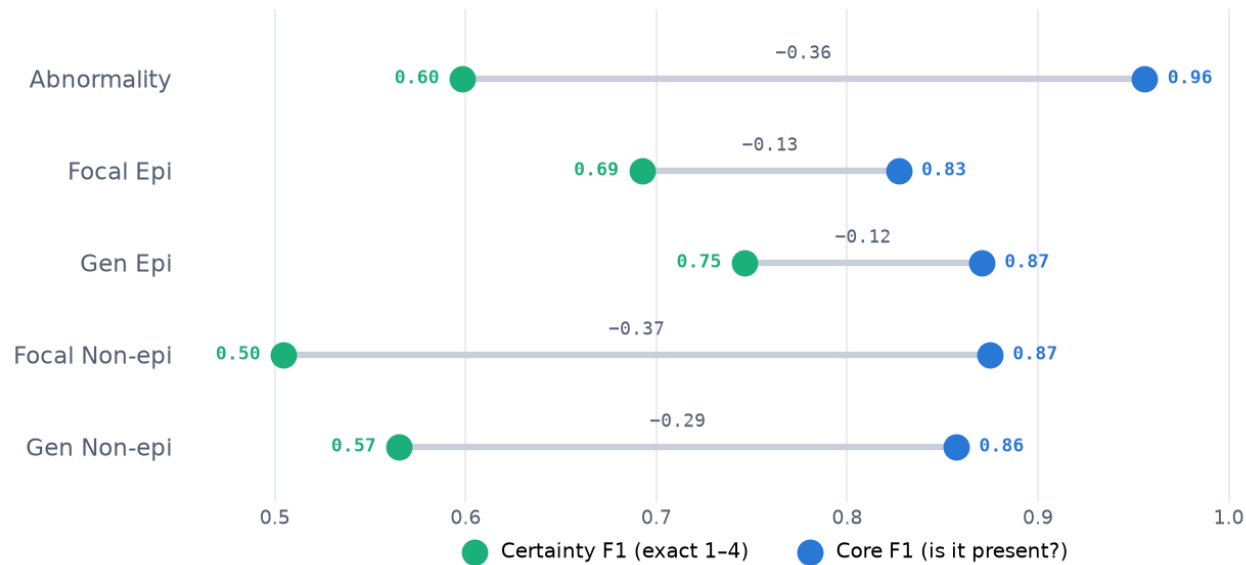
MedGemma-27B · v2 · Q2_K — core vs certainty

F1: binary present/absent vs strict exact-level match · pooled n=1994



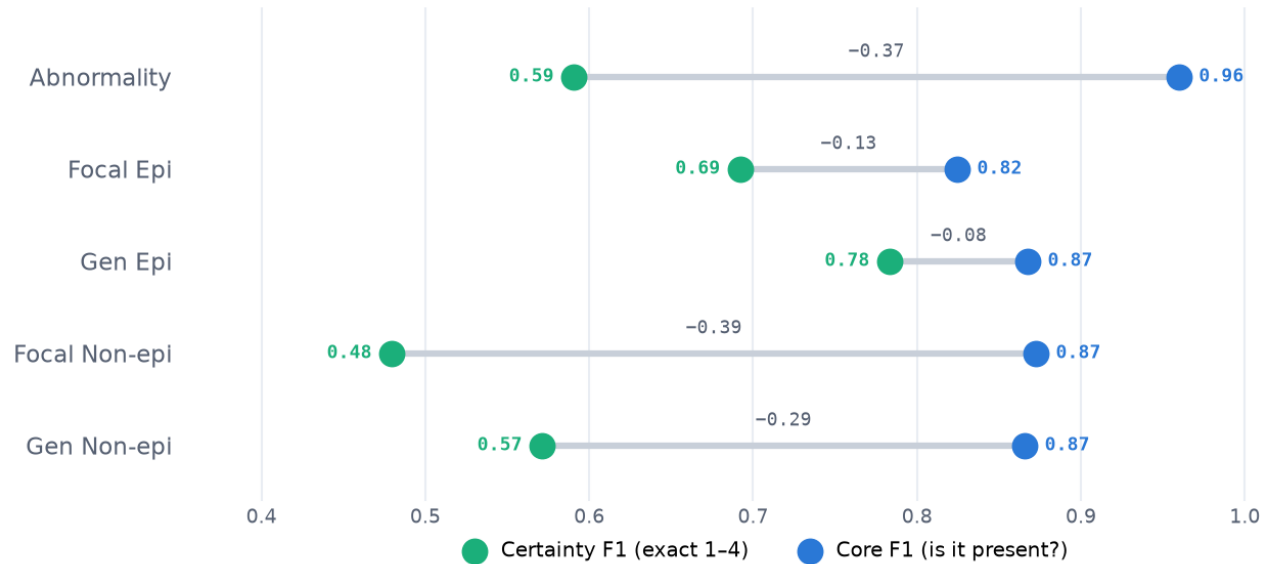
MedGemma-27B · v3 · Q2_K — core vs certainty

F1: binary present/absent vs strict exact-level match · pooled n=1994



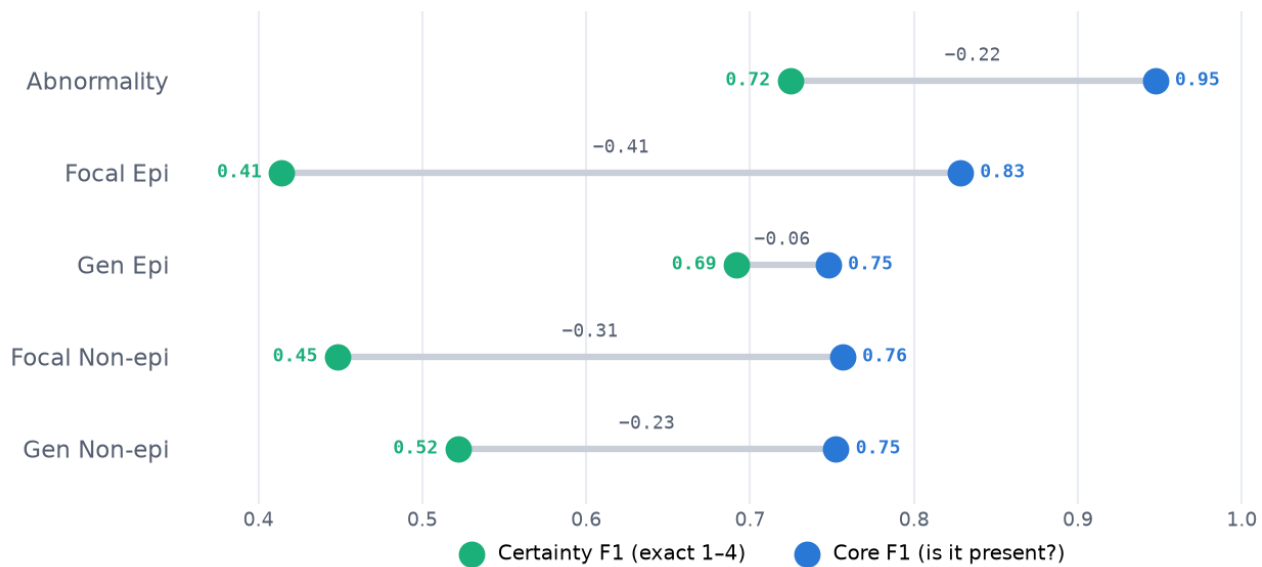
MedGemma-27B · v4 · Q2_K — core vs certainty

F1: binary present/absent vs strict exact-level match · pooled n=1994



Mistral-7B (paper) — core vs certainty

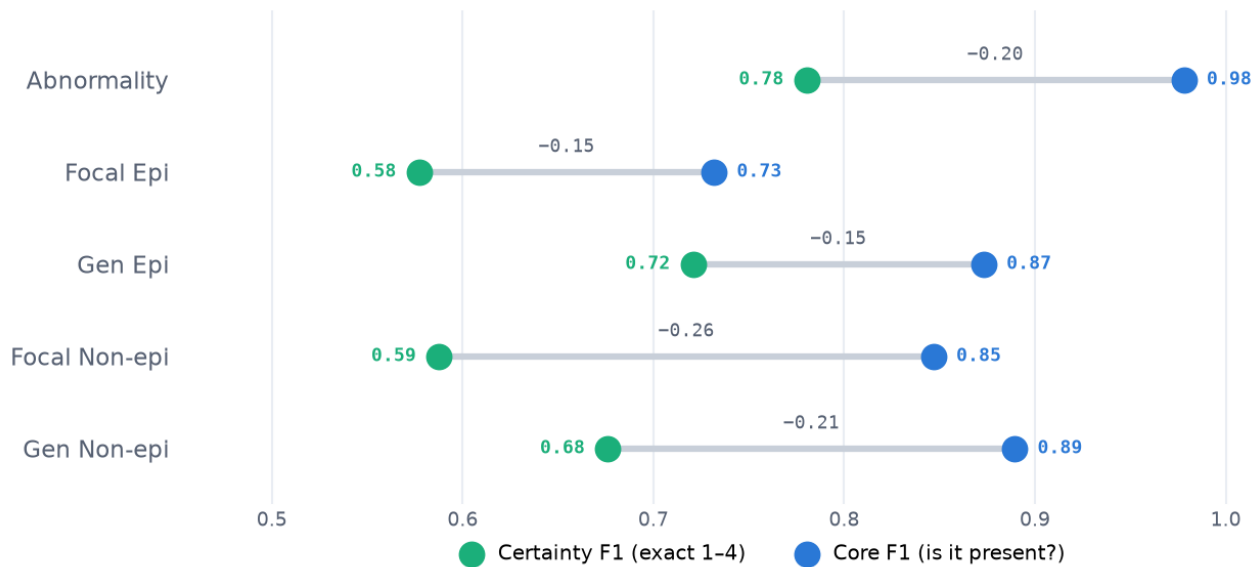
F1: binary present/absent vs strict exact-level match · pooled n=1994



Q4_K_S:

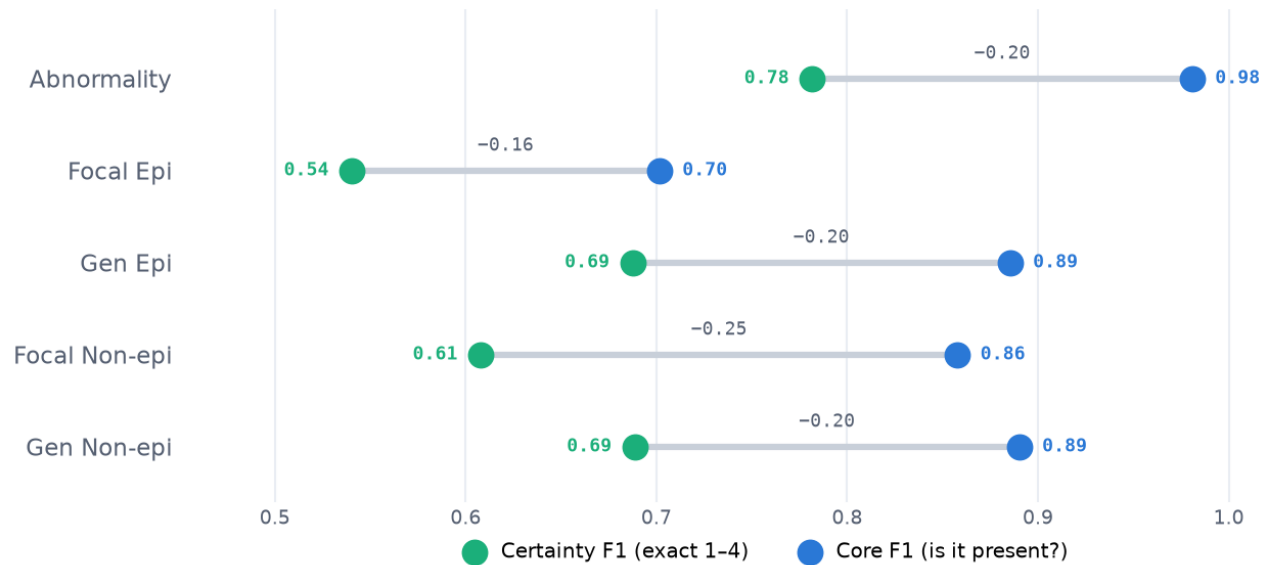
MedGemma-27B · v1 · Q4_K — core vs certainty

F1: binary present/absent vs strict exact-level match · pooled n=1994



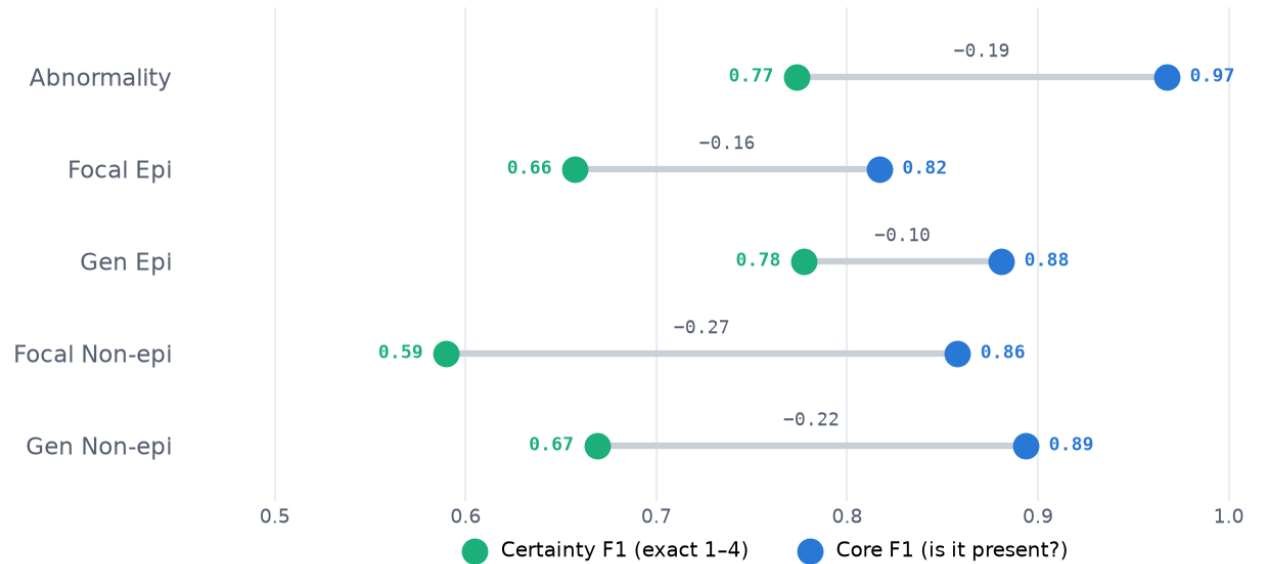
MedGemma-27B · v2 · Q4_K — core vs certainty

F1: binary present/absent vs strict exact-level match · pooled n=1994



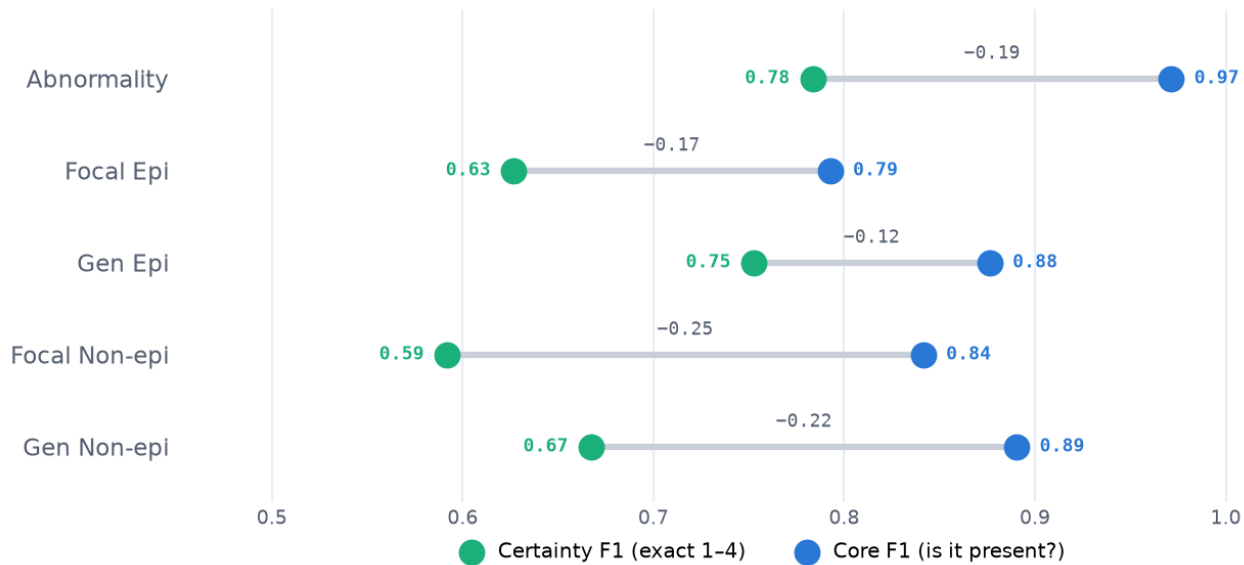
MedGemma-27B · v3 · Q4_K — core vs certainty

F1: binary present/absent vs strict exact-level match · pooled n=1994



MedGemma-27B · v4 · Q4_K — core vs certainty

F1: binary present/absent vs strict exact-level match · pooled n=1994



All numbers — every run, both datasets

Complete Core F1 (vs LD) per category plus fully-correct counts, for all 8 MedGemma runs (4 prompts × 2 quants), the paper's **Mistral-7B**, and the **human** second annotator (SG). Generated by [analysis/make_tables.py](#) from results/*.json, the released Mistral prediction DB, and the sg_labels stored in our JSONs — nothing is hand-typed.

Pooled — Zoe + Maria (n=1994)

Model	Prompt	Quant	Abnormality	Focal Epi	Gen Epi	Focal Non-epi	Gen Non-epi	Fully-correct
MedGemma-27B	v1	Q2_K	0.97	0.80	0.89	0.88	0.84	1672
MedGemma-27B	v1	Q4_K_S	0.98	0.73	0.87	0.85	0.89	1622
MedGemma-27B	v2	Q2_K	0.98	0.64	0.88	0.86	0.82	1576
MedGemma-27B	v2	Q4_K_S	0.98	0.70	0.89	0.86	0.89	1652
MedGemma-27B	v3	Q2_K	0.96	0.83	0.87	0.87	0.86	1679
MedGemma-27B	v3	Q4_K_S	0.97	0.82	0.88	0.86	0.89	1659
MedGemma-27B	v4	Q2_K	0.96	0.82	0.87	0.87	0.87	1662
MedGemma-27B	v4	Q4_K_S	0.97	0.79	0.88	0.84	0.89	1617
Mistral-7B (paper)	—	—	0.95	0.83	0.75	0.76	0.75	1485

Model	Prompt	Quant	Abnormality	Focal Epi	Gen Epi	Focal Non-epi	Gen Non-epi	Fully-correct
<i>Human (SG, 2nd annotator)</i>	—	—	<i>0.98</i>	<i>0.86</i>	<i>0.88</i>	<i>0.91</i>	<i>0.90</i>	<i>1791</i>

Zoe — in-distribution (n=1495)

Model	Prompt	Quant	Abnormality	Focal Epi	Gen Epi	Focal Non-epi	Gen Non-epi	Fully-correct
MedGemma1 27B	v1	Q2_K	0.98	0.77	0.90	0.87	0.85	1255
MedGemma1 27B	v1	Q4_K_S	0.98	0.66	0.88	0.83	0.90	1205
MedGemma2 27B	v2	Q2_K	0.98	0.58	0.89	0.84	0.84	1159
MedGemma2 27B	v2	Q4_K_S	0.98	0.64	0.89	0.84	0.90	1215
MedGemma3 27B	v3	Q2_K	0.97	0.79	0.87	0.88	0.88	1272
MedGemma3 27B	v3	Q4_K_S	0.98	0.78	0.89	0.84	0.91	1240
MedGemma4 27B	v4	Q2_K	0.97	0.79	0.87	0.87	0.89	1253
MedGemma4 27B	v4	Q4_K_S	0.98	0.75	0.89	0.82	0.90	1201
<i>Mistral-7B (paper)</i>	—	—	<i>0.96</i>	<i>0.84</i>	<i>0.73</i>	<i>0.76</i>	<i>0.79</i>	<i>1126</i>
<i>Human (SG, 2nd annotator)</i>	—	—	<i>0.98</i>	<i>0.84</i>	<i>0.89</i>	<i>0.90</i>	<i>0.90</i>	<i>1328</i>

Maria — out-of-distribution (n=499)

Model	Prompt	Quant	Abnormality	Focal Epi	Gen Epi	Focal Non-epi	Gen Non-epi	Fully-correct
MedGemma1 27B	v1	Q2_K	0.92	0.88	0.84	0.91	0.75	417
MedGemma1 27B	v1	Q4_K_S	0.95	0.90	0.84	0.89	0.83	417
MedGemma2 27B	v2	Q2_K	0.97	0.76	0.84	0.91	0.76	417
MedGemma2 27B	v2	Q4_K_S	0.97	0.88	0.86	0.91	0.84	437
MedGemma3 27B	v3	Q2_K	0.91	0.90	0.86	0.87	0.73	407
MedGemma3 27B	v3	Q4_K_S	0.93	0.91	0.84	0.90	0.82	419
MedGemma4 27B	v4	Q2_K	0.92	0.89	0.86	0.89	0.74	409

Model	Prompt	Quant	Abnormality	Focal Epi	Gen Epi	Focal Non-epi	Gen Non-epi	Fully-correct
MedGemma-27B	v4	Q4_K_S	0.94	0.91	0.84	0.90	0.84	416
Mistral-7B (paper)	—	—	0.90	0.81	0.84	0.74	0.54	359
Human (SG, 2nd annotator)	—	—	0.98	0.90	0.84	0.94	0.90	463

Mistral’s per-dataset F1 reproduces the paper’s Table III; the human row is the annotator ceiling (SG vs LD).

Reproduce

```
source .venv/bin/activate
python -m analysis.make_tables           # the numbers tables above
python -m analysis.plot_v34             # variant-comparison charts
python -m analysis.plot_dumbbells       # per-algorithm core-vs-certainty dumbbells
```

a run – prompt / dataset / quant are env-selected:

```
DATASET=zoe GGUF_QUANT=Q2_K PROMPT_VARIANT=v3 CTX_SIZE=8192 \
  sbatch cpu/run_benchmark.sbatch 0 1495 results/zoe_v3_cpu_q2_k_full_n1495.json
```

The baseline analysis (v1/v2 detail, generalization, calibration, over/under-calling, pooled vs-Mistral and vs-human charts) is in [results_baseline.md](#). Model: MedGemma-27B GGUF, llama.cpp grammar-constrained, temp 0, 64-core CPU.